MCQ A · order and repetition

After S25; MCQ B after S35.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

1. A three-digit code chosen from 0, 1, 2, 3, with repetition allowed, gives:

A. 4×3 B. 3^4

C. 4^3 D. $4!$

2. Two distinct roles are given to two people among five. The number of assign is:

A. 10 B. 20

C. 25 D. 120

3. The number of permutations of four distinct objects is:

A. 16 B. 12

C. 4 D. 24

4. For two disjoint finite sets A, B, $|A \cup B|$ is:

A. $|A| + |B|$ B. $|A||B|$

C. $|A| - |B|$ D. $|A \cap B|$

5. With two cards A and B, the lists of length 2 with repetition are:

A. AB, BA B. AA, BB

C. AA, AB, BA, BB D. A, B

MCQ B · complement and double counting

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

6. Choosing three pupils among eight with no assigned role is counted by:

A. 8^3 B. $\binom{8}{3}$

C. $8 \times 7 \times 6$ D. 3^8

7. The number of subsets of a five-element set is:

A. $5!$ B. 10

C. 25 D. 32

8. Pascal's rule gives $\binom{6}{3}$ equal to:

A. $\binom{5}{2}+\binom{5}{3}$ B. $\binom{5}{2}\binom{5}{3}$

C. $2\binom{6}{2}$ D. $\binom{6}{4}+1$

9. An urn contains 3 red and 2 blue distinct balls. The number of pairs without replacement containing at least one red ball is:

A. 6 B. 10

C. 9 D. 3

10. Six people are split into two numbered groups of three. The number of ways is:

A. 10 B. 20

C. 40 D. 6!

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. $\binom{n}{0}=0$.

2. Counting with the complement can avoid distinguishing several cases.

3. Drawing three objects simultaneously always gives the objects an order.